Paper Club · Sep 9

Stakeholder

18 + 3 minutes

Leaderboard · stack · motion-control spine

arXiv 2512.06951 · 1st Place BEHAVIOR 2025

Seat 5

Why this matters

50 long-horizon household tasks · partial-progress q_score

1st Place · BEHAVIOR 2025 Challenge

Robot Learning Collective · arXiv 2512.06951

q_score	binary success
~26%	**~11–12%**
pub 0.2605 · priv 0.2599	partial ≈ half of q-score

Stack overview

- **Pi0.5 / OpenPI**
- **Correlated FM β=0.5**
- **Mixed-layer KV**
- **System 2 voting**
- **Task embeddings**
- **4 FT ckpts**
- **Gripper heuristics**

Motion / Control Spine · Not classical planning

Learned action-chunk FM + rolling receding-horizon + stage hierarchy + rule recovery

Predict
H = 30
flow-matching action chunk

→

Execute
26 steps
mostly open-loop inside chunk

→

Soft inpaint
4 steps
$t > 0.3$ only
corr-aware

→

Replan
every 26
rolling RTC-style stitch

Closed-loop overrides (outer loop)

System 2 · Stage Voting

5–15 stages / task

vote window = 3 (≥2/3 forward)

rollback rules for aliasing

NON-MARKOVIAN context

Gripper Recovery Heuristic

force open if closed in stage

never closed in demos

paper: ~2.2× q on 13×3 eps

CLOSED-LOOP OVERRIDE

Method Highlights

Correlated-noise FM

$\varepsilon \sim N(0, \beta\Sigma+(1-\beta)I)$
$\beta=0.5 \cdot \Sigma$ = action cov ($H \cdot D=690$)
Aligns train ↔ soft inpaint

Mixed-layer KV attn

Learnable combo all VLM KV
→ each action-expert layer
Init = identity (Pi0.5)

Task embeddings

Drop language prompts
50 trainable 2048-D embs
Ship 4 group FT checkpoints

Train extras

Multi-sample $N=15$
FAST aux $\lambda=0.05$
Cubic 1.3× compression

System 2 · Non-Markovian Stage Voting

t−2	t−1	t	VOTE
stage pred $\hat{s}$	stage pred $\hat{s}$	stage pred $\hat{s}$ →	≥2/3 forward

Forward

Majority agrees
on next stage
→ advance

Hold / Skip

Ambiguous vote
→ stay or skip
per rules

Rollback

Radio / aliasing
→ stage rollback
(visual confusion)

Key Findings

Gripper heuristic

~2.2× q on grasp subsets
(13 tasks × 3 eps)

Emergent recovery

Multi-task IL → recovery
behaviors (qualitative)

Undertrained

~2 epochs only
~$13k · 8×H200

Cheap infer

Single RTX 4090 24GB
pub ≈ priv generalization

Task adaptation of Vision-Language-Action model: 1st Place Solution for the 2025 BEHAVIOR Challenge

Ilia Larchenko*
ilya.larchenko@gmail.com

Gleb Zarin*
zaringleb@gmail.com

Akash Karnatak*
akashkarnatak.pro@gmail.com

*Independent Researchers

December 7, 2025

Abstract

We present a vision-action policy that won 1st place in the 2025 BEHAVIOR Challenge—a large-scale benchmark featuring 50 diverse long-horizon household tasks in photo-realistic simulation, requiring bimanual manipulation, navigation, and context-aware decision making.

Building on the Pi0.5 architecture, we introduce several innovations. Our primary contribution is **correlated noise for flow matching**, which improves training efficiency and enables **correlation-aware inpainting** for smooth action sequences. We also apply **learnable mixed-layer attention** and **System 2 stage tracking** for ambiguity resolution. Training employs **multi-sample flow matching** to reduce variance, while inference uses **action compression** and challenge-specific **correction rules**.

Our approach achieves 26% q-score across all 50 tasks on both public and private leaderboards.

1 Introduction

1.1 The BEHAVIOR Challenge

The BEHAVIOR Challenge presents a demanding benchmark for embodied AI: performing 50 diverse, long-horizon household tasks in photo-realistic simulation. Tasks range from simple (turning on radio) to complex multi-step activities (cooking a hotdog). The challenge requires:

- **Long-horizon execution**: 6.6 minutes on average per task, with the longest tasks averaging 14 minutes
- **Bimanual manipulation**: Coordinated use of two 7-DOF arms with parallel-jaw grippers
- **Mobile navigation**: Indoor navigation through cluttered environments
- **Multi-camera perception**: Processing RGB images from head and both wrist cameras
- **Task diversity**: 50 different activities evaluated with a single policy (or small set of checkpoints)

The benchmark uses OmniGibson simulation built on NVIDIA Isaac Sim, providing realistic physics and rendering. Each task is evaluated over 10 episodes with randomized initial conditions, and performance is measured by a q-score that combines success rate with partial credit for subtask completion.

1.2 Key Challenges

Long-horizon household manipulation poses several fundamental challenges:

- **Compounding errors.** With episodes spanning thousands of timesteps, small prediction errors can accumulate. This demands either extremely accurate predictions or robust recovery behaviors.
- **Non-Markovian states.** Many task states are visually ambiguous—the robot holding a radio at the start of a task looks identical to holding it at the end. Without memory of past actions or explicit stage tracking, the policy cannot distinguish these states and may execute incorrect actions.
- **No recovery demonstrations.** The training data consists of successful demonstrations only. When the robot deviates from demonstrated trajectories (inevitable given compounding errors), it encounters states never seen during training. The policy must somehow generalize to recover from these out-of-distribution situations.
- **Multi-modal action distributions.** Many states admit multiple valid action sequences (e.g., which hand to use, which object to grasp first). Different episodes of the same tasks were completed at different speeds in the training data.

1.3 Our Approach

We build upon Pi0.5 [3], a vision-language-action (VLA) model that uses flow matching to predict action sequences.

Paper title page · arXiv 2512.06951

Cost & handoff

~$13k train · 4090 infer · Reviewer next, Empiricist owns ablations

Questions?

Leaderboard · deploy · cost — ablations → Empiricist